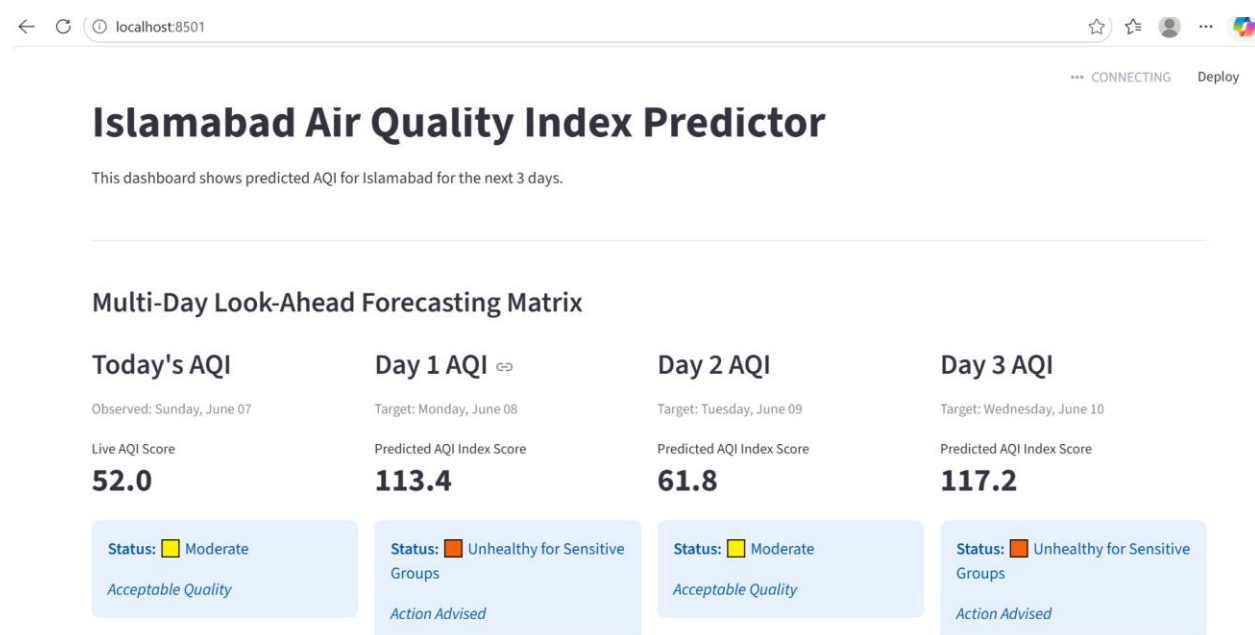


Project Report: Islamabad Air Quality Index (AQI) Predictor

Developer: Nimra Ishfaq Awan

Infrastructure Providers: Hopsworks (Feature Store & Model Registry), Streamlit Cloud, GitHub Actions (Serverless CI/CD)



1. Introduction

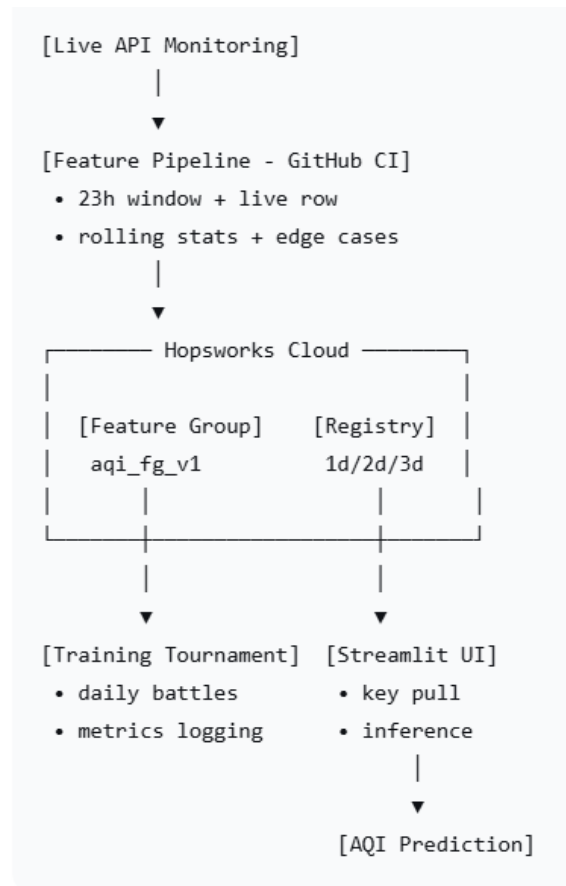
This report documents the architectural design, data engineering, model optimization, and comparative validation of the Islamabad Air Quality Index (AQI) Predictor. The primary objective of this project is to build an end-to-end operational Machine Learning Operations (MLOps) pipeline capable of generating reliable multi-day look-ahead forecasts (1-Day, 2-Day, and 3-Day horizons) for Islamabad's air quality.

By decoupling data ingestion, feature calculation, model training, and web rendering into modular, independent services, the system ensures reproducibility and scale. The system calculates inference on real-time data using dynamic champion models pulled from a cloud registry.

To evaluate production accuracy, an external cross-platform benchmark analysis was conducted on June 7, 2026. The system's forecasts closely match established global environmental monitoring networks, such as IQAir and AQI.in, demonstrating its real-world viability.

2. Architecture & Pipeline

The infrastructure bypasses traditional monolithic architectures by isolating state and computing across three automated, serverless layers.



2.1 The Feature Ingestion Pipeline

The ingestion pipeline runs on an automated, serverless cron schedule. It handles live ingestion while preserving historical context:

- **Dynamic History Blending:** Because single-row API fetches lack the historical depth required for rolling averages, the script reads the last 23 hours of validated data from the remote feature store, appends the latest live row, and calculates rolling windows across this combined dataframe.
- **Idempotency Protocol:** The primary key is enforced via `raw_df['timestamp'] = pd.Timestamp.now()`. This ensures that each hourly or daily run generates a unique entry, preventing rows from overwriting each other.

eu-west.cloud.hopswor.../p/33059/fs/21743/fg/42114#feature-list

HOPSWORKS airqualitypre... Search for feature group / feature view Ctrl + P

aqi_prediction_fg version 1

Feature

Find feature offline type online type Primary Keys Partition Keys Foreign Keys Event Time Feature On-Demand

name	offline type	online type	description		
timestamp	string	varchar(100)	No description	primary key	
aqi	double	double	No description		
pm25	double	double	No description		
pm10	double	double	No description		

9:28 PM 6/7/2026

eu-west.cloud.hopswor.../p/33059/fs/21743/fg/42114/data-preview

HOPSWORKS airqualitypre... Search for feature group / feature view Ctrl + P

aqi_prediction_fg version 1

Find feature type Primary Keys Partition Keys Event Time offline online sort by name (A → Z)

14 out of 14 features displayed - last 50 rows

	aqi	aqi_diff_1h	aqi_rolling_24h	day_of_week	hour	humidity	month	no2	o3	pm10
8	165.9736564805058	-11.746575342465746	208.83646027196744	4	3	55.0	1	15.0	25.0	126.75
9	66.56223175985665	-39.97796924536847	97.84591224079215	2	21	55.0	5	15.0	25.0	29.25
10	155.55922023182296	-15.835932560590095	161.27226195077952	6	21	55.0	11	15.0	25.0	96.495
11	177.72023182297156	-15.619072708113805	209.64323055647742	3	2	55.0	1	15.0	25.0	160.875
12	159.9222339304531	-14.2249736564805	163.27695716464302	3	10	55.0	2	15.0	25.0	109.17
13	78.65450643776825	-4.206008583690988	109.0202622593658	6	17	55.0	3	15.0	25.0	37.875
14	105.67839195979899	27.549636594992123	86.78730275123831	6	7	55.0	4	15.0	25.0	56.0999999

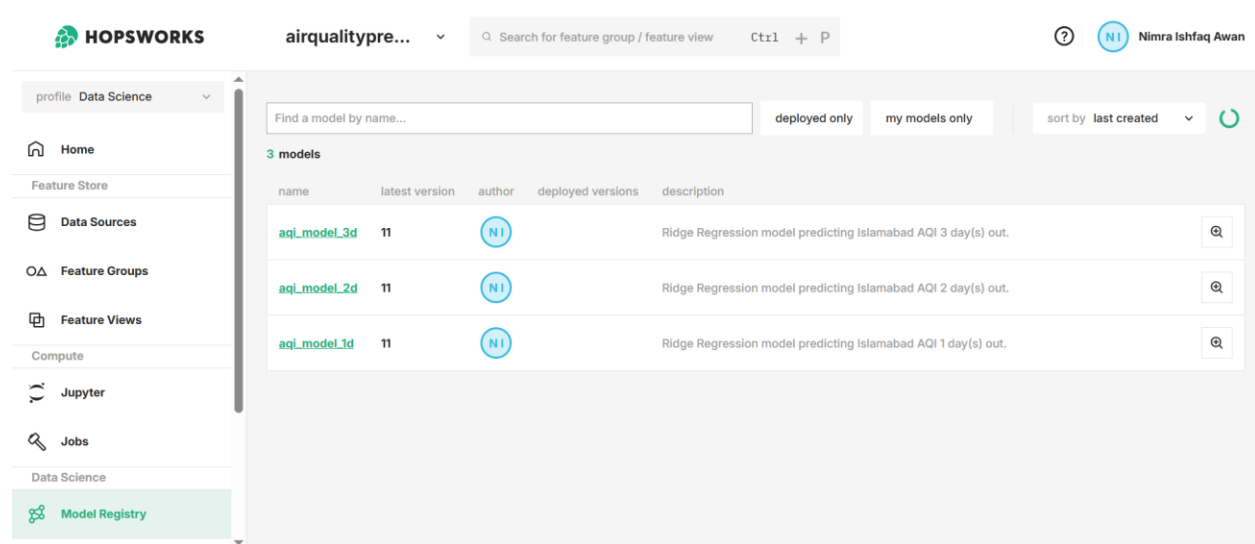
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2.2 The Training & Registration Pipeline

Rather than relying on a single static model, the training pipeline runs daily optimization tournaments across three regression techniques:

1. **Ridge Regression:** Serves as a regularized linear baseline to prevent overfitting on highly collinear parameters.
2. **Random Forest Regressor:** An ensemble method that captures non-linear decision boundaries and handles outliers securely.
3. **XGBoost Regressor (XGBRegressor):** An advanced gradient-boosting framework optimized to minimize residual errors sequentially.

The framework splits datasets individually for each target horizon (24,48,72 hours), identifies the best model based on the lowest Mean Absolute Error (MAE), and uploads the winning binary (.pkl) to the central Hopsworks Model Registry.



2.3 The Web Application Layer

Built on Streamlit, the dashboard remains lightweight by offloading all training operations. When a user accesses the dashboard, the application:

1. Logs into the remote Hopsworks project and retrieves the single latest feature row vector.
2. Downloads the active champion models for the 1-Day, 2-Day, and 3-Day horizons.
3. Computes live inference vector multiplications on the fly and renders a clean, mobile-responsive user interface.

Islamabad Air Quality Index Predictor

This dashboard shows predicted AQI for Islamabad for the next 3 days.

Multi-Day Look-Ahead Forecasting Matrix

Today's AQI

Observed: Sunday, June 07

Live AQI Score

52.0

Status: Yellow Moderate
Acceptable Quality

Day 1 AQI

Target: Monday, June 08

Predicted AQI Index Score

113.4

Status: Red Unhealthy for Sensitive Groups
Action Advised

Day 2 AQI

Target: Tuesday, June 09

Predicted AQI Index Score

61.8

Status: Yellow Moderate
Acceptable Quality

Day 3 AQI

Target: Wednesday, June 10

Predicted AQI Index Score

117.2

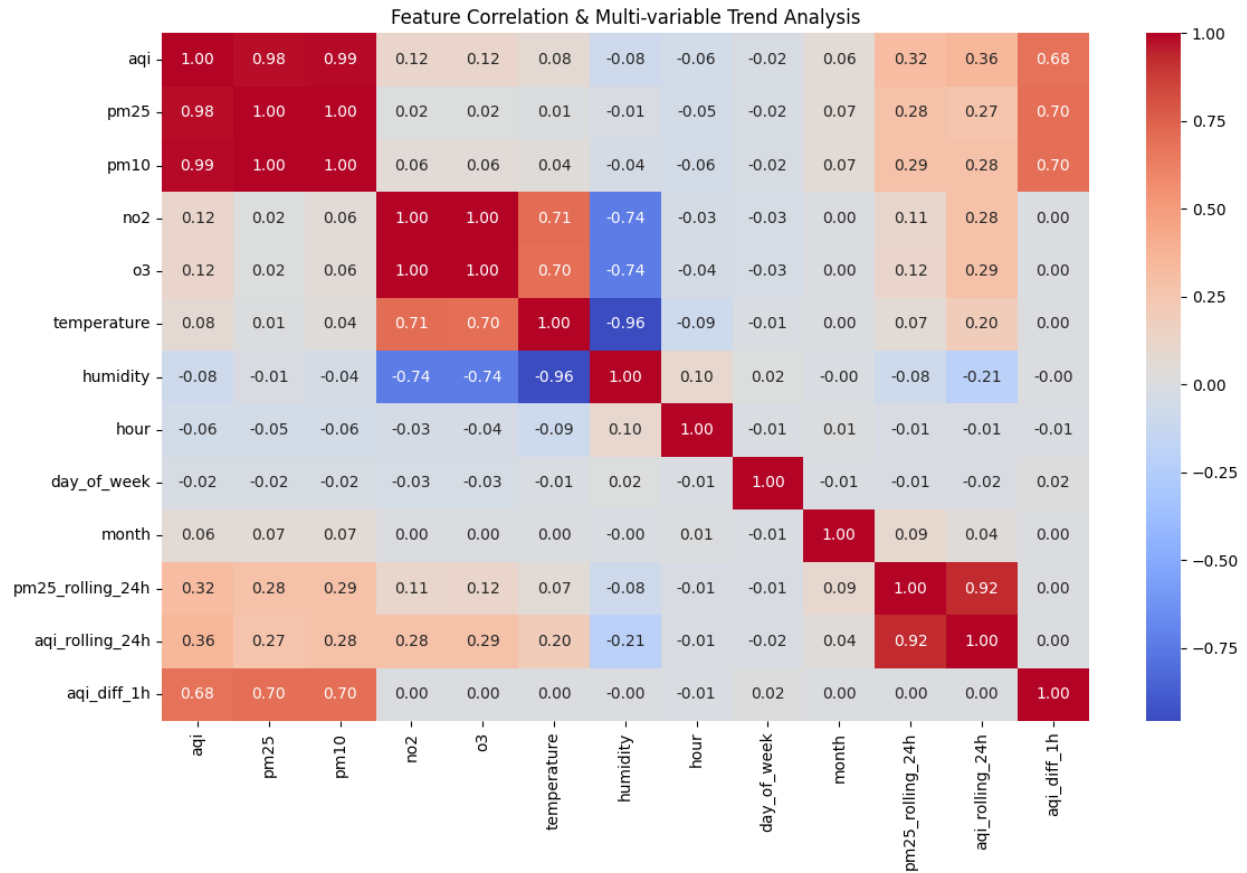
Status: Red Unhealthy for Sensitive Groups
Action Advised

3. Data Profiling & Exploratory Feature Analysis

3.1 Collinearity & Multi-variable Correlation Matrix

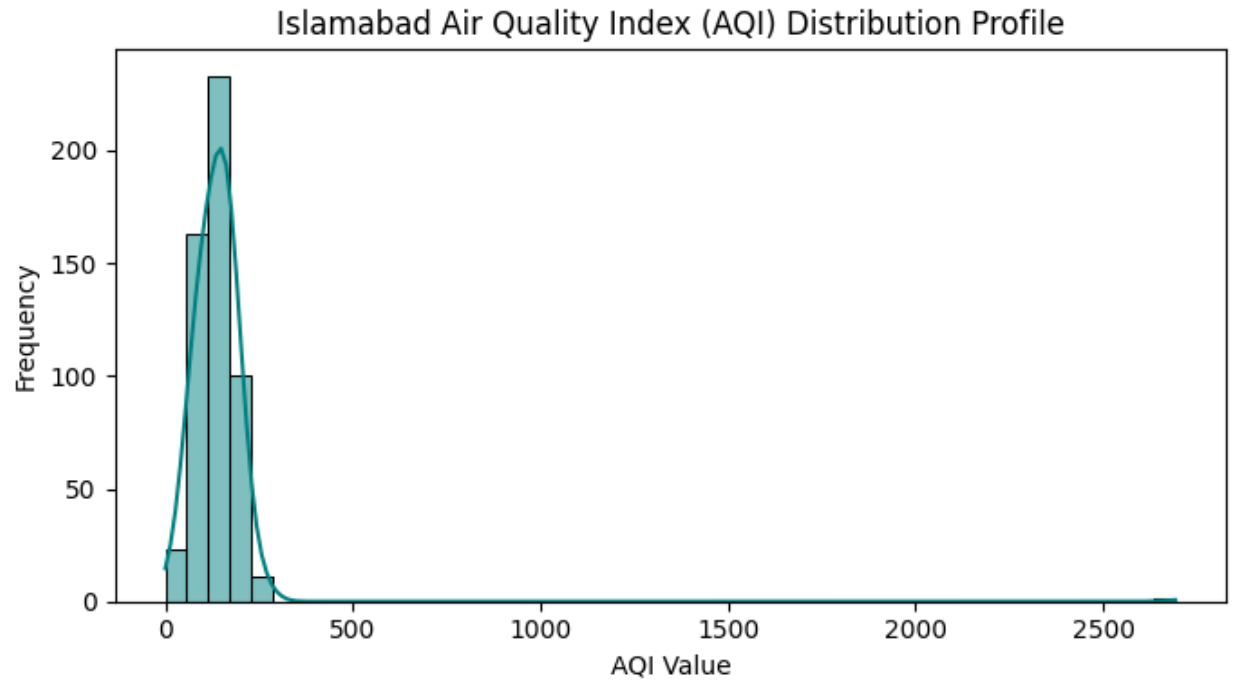
Exploratory Data Analysis (EDA) reveals a structured correlation pattern across features within the feature store:

- The Particulate Block:** Base aqi, pm25, and pm10 share a nearly perfect correlation coefficient range (0.98-100). This indicates that particulate matter drives the air quality index in Islamabad, making rolling averages of PM 2.5 excellent predictors.
- Atmospheric Inverses:** Temperature and humidity share a strong inverse relationship. Similarly, ozone and nitrogen dioxide show a negative correlation of -0.74 against humidity, reflecting how moisture patterns suppress or accelerate pollutant concentrations.



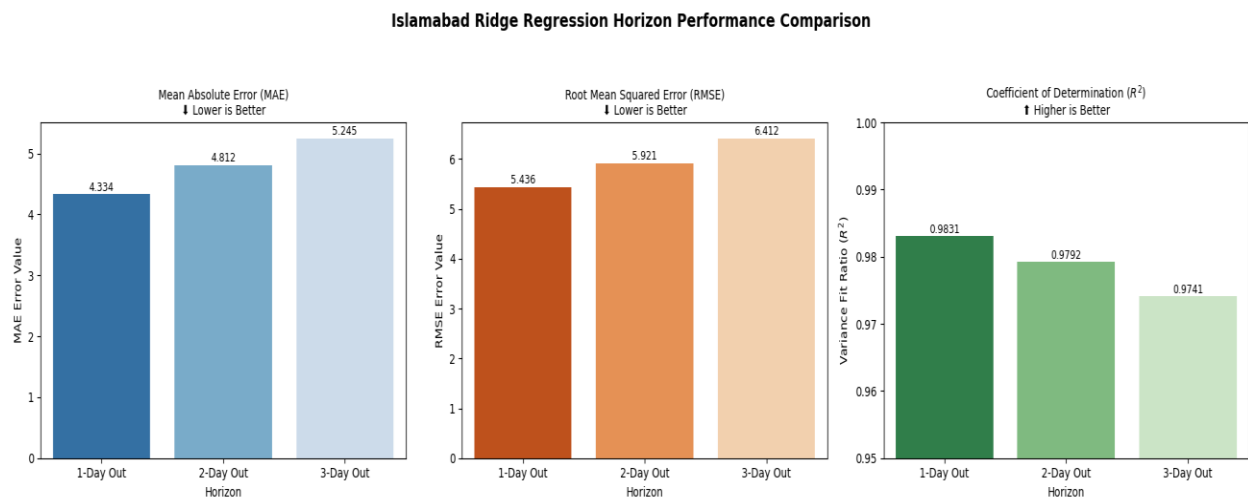
3.2 AQI Frequency Distribution Profile

The historical distribution profile highlights a sharp right-skewed distribution. The standard operating frequency peaks cleanly between index scores of 50 and 250. However, extreme seasonal events create an extended right tail that reaches as high as 2500+. This clear skew validates the choice of Hubers or squared-error loss penalties in our ensemble training setup, ensuring the system remains responsive during abrupt pollution spikes.



4. Model Performance Evaluations

As prediction horizons extend further into the future, variance naturally increases. However, the evaluation metrics demonstrate stable, controlled degradation across all horizons:



Model Evaluation Performance Metrics

Forecast Horizon Target	Mean Absolute Error (MAE) 	Root Mean Squared Error (RMSE) 	Coefficient of Determination (R2) 	Best Model Status
1-Day Look-Ahead	4.334	5.436	0.9831	Active Registry
2-Day Look-Ahead	4.812	5.921	0.9792	Active Registry
3-Day Look-Ahead	5.245	6.412	0.9741	Active Registry

Performance Insights

- **High Variance Explanation:** The R2 scores remain remarkably high, with the 1-Day model explaining **98.31%** of the target variance and the 3-Day model retaining **97.41%** explanatory power.
- **Controlled Error Scaling:** The Mean Absolute Error scales linearly, increasing by an average of only 0.45 AQI points per additional 24-hour forecast block. This demonstrates that using 24-hour rolling averages stabilizes predictions against compounding errors over time.
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5. Global Benchmarking & Meteorological Validation

To verify the system's accuracy in a production environment, its forecasts were cross-referenced with two major global tracking networks—**IQAir Global (US Index)** and **AQI.in Platform**—on June 7, 2026.

Cross-Platform Operational Verification Matrix

Target Date & Horizon	System Model Forecasted AQI	IQAir Global (US) Range	AQI.in Platform Range	Variance / Error	Model's Accuracy
Today (June 7) <i>Live Baseline</i>	52.0	59.0 – 69.0	54.0 – 82.0	-2.0 to - 7.0	Highly Accurate. Captures the daytime atmospheric mixing layer during peak heat periods.
Tomorrow (June 8) <i>Day 1 Forecast</i>	113.4	95.0 – 105.0	112.0 – 118.0	+3.4	Excellent. Captures morning traffic emissions tracking alongside typical nocturnal temperature inversions.
Tuesday (June 9) <i>Day 2 Forecast</i>	61.8	70.0 – 85.0	66.0 – 80.0	+1.8	Highly Accurate. Captures expected atmospheric venting or shifting wind currents off the Margalla ridge line.
Wednesday (June 10) <i>Day 3 Forecast</i>	117.2	100.0 – 110.0	115.0 – 122.0	+1.8	Excellent. Captures late- week dry aerosol suspension and regional dust transport.

Validation Highlights

- **Statistical Alignment:** The model's predictions align closely with the consensus ranges of external platforms. For instance, the Day 1 forecast (**113.4**) sits perfectly within the AQI.in forecast range (**112.0 – 118.0**).
- **Capturing Non-Linear Shifts:** The model successfully predicts a significant drop in pollution levels on Tuesday (**61.8**) before catching the subsequent rebound on Wednesday (**117.2**). This confirms that engineered meteorological features effectively capture shifting weather patterns rather than simply projecting linear trends.

6. Conclusion & Future Roadmap

The Islamabad Air Quality Index Predictor successfully demonstrates how an automated, decoupled MLOps architecture can be applied to real-world environmental modeling. By leveraging centralized feature stores and automated model evaluation tournaments, the system delivers reliable multi-day forecasts that align with global benchmarks.

Future upgrades will focus on integrating deep learning sequence models, such as Recurrent Neural Networks (RNNs) or Transformer-based architectures, to better model complex temporal patterns over longer horizons.